

Kun Zhang

PhD Applicant — Large Language Models: Post-Training, Alignment, and Evaluation

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My work is on large language models, across evaluation validity, mechanistic interpretability, alignment, and distribution-free guarantees. What holds it together is the experimental design rather than the topic. Before I report a positive result I build the control that would produce it for the wrong reason, and where that control is not rejected I report the smallest effect the test could have detected. **Several of the papers below end in a claimed effect that does not survive that test.**

Education

University of Leeds

Leeds, United Kingdom

MSc Advanced Computer Science — taught and assessed entirely in English

Sep 2025 – Sep 2026

- Dissertation: a leakage-controlled audit of whether financial text adds forecasting value beyond price and firm-level baselines.
- Selected module marks: Deep Learning **84** • Knowledge Representation & Reasoning **81** • Data Mining & Text Analytics **81** • Machine Learning **77**

Hunan University of Finance and Economics

Changsha, China

BSc Computer Science and Technology

Sep 2021 – Jun 2025

- Final-year project: GAN-based face image inpainting

Publications

All papers below are **first-authored**; four accepted, three under review. Papers marked † were co-authored with my MSc supervisor Dr Chunwei Xia as corresponding author; the remaining two are **sole-authored**.

ACCEPTED

- [1] K. Zhang, C. Xia†. *Same Exam, Different Scores: A Tokenizer-Grounded Audit of Evaluation-Protocol Sensitivity on Chinese Benchmarks*. **NLPCC 2026** (Resources and Evaluation track) — **Oral**, Springer LNCS.

Pre-registered a mechanistic prediction that first-token-argmax scoring makes the same benchmark assign systematically different scores to the same model. The predicted effect held; the reported ranking is an artefact of the protocol.

- [2] K. Zhang, C. Xia†. *A Point-in-Time Audit of Graph-Snapshot Leakage in GNN Stock Selection*. **PRICAI 2026** — regular paper (12–16 pp.), **Oral**, Springer LNAI.

Rebuilt institutional co-holding graphs from raw SEC 13F filings without look-ahead. Most reported gains from graph-snapshot methods do not survive the correction.

- [3] K. Zhang. *PCP: A Controlled Benchmark for Pragmatic Intention Recognition in Clinical Communication*. **LUHME 2026 @ EMNLP** — **Oral**. **Sole author**.

A 3,533-item benchmark for recovering intended meaning from hedging, indirectness and implicature. Eleven systems sit near ceiling on the controlled split (98.7–99.8%) but fall to 79.7–90.6% on externally grounded cases, so a broad controlled probe can hide the difficulty that close distractors expose.

- [4] K. Zhang. *Grounding or Lexical Overlap? A Matched-Pair Study of Internals-Based Context-Faithfulness Probes*. **GroundLM 2026 @ EMNLP** — **Findings**. **Sole author**.

Faithfulness scores from internals-based probes track lexical overlap at $r = 0.99$; on matched pairs the signal halves, so what the probes measure is largely surface overlap rather than grounding.

UNDER REVIEW

Three further first-author manuscripts are under review; titles and venues are withheld during anonymous review.

Research Themes

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- **Evaluation validity and leakage control** [1] [2] [3] — when a reported score measures the protocol, a look-ahead or a controlled split rather than the capability.
 - **Mechanistic interpretability** [4] — causal interventions on model internals, each paired with a control that would reproduce the observation for a different reason.

- **Alignment and safety** [under review] — which properties a post-training run installs, which it only appears to, and what evidence tells the two apart.
- **Distribution-free guarantees** [under review] — error-rate control without distributional assumptions.

Industry Experience

Hunan Chaote Network Technology

AI Application Development (intern) — built a retrieval-augmented generation architecture

Changsha, China

Nov 2024 – Jun 2025

Guangdong United Information Technology

Data Analysis (intern) — business data evaluation

China

Feb 2024 – Aug 2024

Academic Service

Reviewer, NLPCC 2026 (main conference) • **Reviewer**, GroundLM 2026 @ EMNLP

2026

Awards

- Apr 2024 **National First Prize**, 2nd “Waiwen Award” National College English Vocabulary Competition — **English Majors Division**. China Foreign Languages Publishing Administration.
Awarded in the English Majors division as a Computer Science student.
- Jul 2024 **Second Prize (national)**, 7th National College Students’ Arts Festival — Ensemble Performance, Group A. **Ministry of Education of the PRC**.
- Jun 2024 **First Prize** (preliminary round), 8th National College Business English Competition.
- 2023–24 **Second-Class Scholarship**, Hunan University of Finance and Economics.
- May 2024 **Second Prize** (preliminary round), 5th National Mathematics Ability Challenge.
- Jun 2023 **First Prize** (preliminary round), Global IoT Conference “Black Technology” Competition, University Division.

Competitions

Kaggle Competitions Expert — ranked **2,095 of 212,340** (top 1%); 1 silver and 1 bronze medal.

Technical

Frameworks PyTorch, HuggingFace Transformers, scikit-learn, PyTorch Geometric

Referees and their contact details are available on request.